Lesson 3.1: ETL pipelines, data warehousing and indexing

What you learn

- **ETL.** Extract the raw data, transform it (clean, dedupe, type), load it into a table. The tested code below does all three.
- **Warehouses.** BigQuery and Snowflake store large data for analysis; they separate storage from compute and charge differently, so check current pricing.
- **Indexing.** An index speeds lookups on a column but slows writes slightly.

Tools and resources (official sites, not checked live)

- BigQuery
- Snowflake

ASSIGNMENT

Practice for lesson 3.1

Business problem: Run a tiny ETL.

Steps

1. Run the tested pipeline.
2. Add a bad row.
3. Check it is dropped.

Expected output: A clean table with one total per city.

SOLUTION (ANSWER KEY)

PYTHON

```
import sqlite3, csv, io
raw = "order_id,city,amount\n1, pune ,500\n2,Delhi,\n3,PUNE,300\n3,PUNE,300\n"
def extract(text): return list(csv.DictReader(io.StringIO(text)))
def transform(rows):
    seen, out = set(), []
    for r in rows:
        if not r["amount"].strip(): continue              # drop rows with a missing amount
        key = r["order_id"]
        if key in seen: continue                          # drop duplicate order ids
        seen.add(key); out.append((int(key), r["city"].strip().title(), int(r["amount"])))
    return out
def load(rows, db=":memory:"):
    con = sqlite3.connect(db); con.execute("create table orders(order_id int primary key, city
text, amount int)")
    con.executemany("insert into orders values (?,?,?)", rows); con.execute("create index
idx_city on orders(city)"); return con
```

The blank-amount row and the duplicate order are removed, and 'pune' becomes 'Pune'.

Check yourself

1. What does the transform step do?
   Answer: Cleans and reshapes data.
2. What does an index cost?
   Answer: Slightly slower writes.

Lesson 3.1 of the **Data Analytics and Data Science** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.